

LOCAL CACCIOPPOLI ESTIMATES AND ROUGH-CORE EXCLUSION

ABSTRACT. We prove the local Caccioppoli estimate needed in the local CKN Type II analysis. The estimate is applied to the local scale-translation charts and pressure decompositions supplied by the representation paper. Together with compact-cylinder CKN bounds and bounded local modulation coefficients, it controls the local suitable package on smaller cylinders and gives windowed H^1 control on bounded cores. Consequently a rough core is not an independent Type II assumption: it either satisfies the local Caccioppoli bounds or enters the cascade/multibubble decomposition. This paper uses the classical Navier–Stokes theory for suitable weak solutions (Leray [Ler34], Caffarelli–Kohn–Nirenberg [CKN82], Scheffer [Sch76], and Lin [Lin98]).

1. RENORMALIZED LOCAL ENERGY INEQUALITY

Let (V, P, a, b) solve the locally repaired renormalized equation on compact cylinders. The local energy inequality is the physical local energy inequality pulled back by the absolutely continuous local chart. It contains the usual Navier–Stokes terms and lower-order terms involving a and b .

2. CACCIOPPOLI ESTIMATE

Theorem 2.1 (Compact-cylinder Caccioppoli estimate). *Let $Q_{2R,I} = B_{2R} \times I$ be compactly contained in the local chart supplied by the representation theorem, and let $I' \Subset I$. For represented Type II branches, the pulled-back local energy inequality on $Q_{2R,I}$ gives*

$$\int_I \int_{B_{2R}} \left(|V|^3 + |P - (P)_{B_{2R}}(\tau)|^{3/2} \right) < \infty,$$

with bounded modulation coefficients a, b on I . Consequently

$$\int_{I'} \int_{B_R} |\nabla V|^2 \leq C \left(1 + \int_I \int_{B_{2R}} |V|^3 + \int_I \int_{B_{2R}} |P - (P)_{B_{2R}}|^{3/2} \right),$$

where C depends on R , $I' \Subset I$, the viscosity, the cutoff choice, and $\|a\|_{L^\infty(I)} + \|b\|_{L^\infty(I)}$.

Proof. Insert a cutoff $\phi(y, \tau) = \eta(\tau)\zeta(y)^2$ in the local energy inequality, with $\zeta = 1$ on B_R and $\text{supp } \zeta \subset B_{2R}$. The convection and cutoff terms are bounded by the displayed local L^3 integral and cutoff constants. The pressure term is estimated after subtracting the spatial mean on B_{2R} . The lower-order modulation terms are bounded by $\|a\|_{L^\infty(I)} + \|b\|_{L^\infty(I)}$, the cutoff constants, and the local L^2 mass. The local L^2 mass is controlled on compact balls by the local L^3 bound. The interior gradient term is absorbed into the left-hand side after the usual Caccioppoli rearrangement. $\square$

Corollary 2.2 (Rough-core alternative). *On unit windows where the local modulation coefficients are uniformly bounded, Theorem 2.1 gives uniform local windowed H^1 control on every bounded core. If the compact-cylinder suitable package fails instead, then the local Caccioppoli estimate implies failure of the local $C + D$ bound on a comparable cylinder. Thus rough-core behavior belongs to the local single-core criterion or to the local multibubble/cascade analysis.*

Proof. The first statement is Theorem 2.1 applied on consecutive windows. For the second, argue by contrapositive: if the local $C + D$ quantities and the modulation coefficients are bounded on a slightly larger cylinder, then Theorem 2.1 bounds E on a smaller cylinder; the compact-ball L^3

bound also controls the local L^2 mass, hence A . Therefore failure of the suitable package must come from local CKN concentration on a comparable cylinder. $\square$

REFERENCES

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